INFORMATION RETRIEVAL ASSIGNMENT

FYP-Based Implementation and Presentation

INDEX
RETRIEVE
RANK
EVALUATE

Presentation Dates: Monday, 12 October 2026 & Tuesday, 13 October 2026

Marks: 10

Mode: Group-based Presentation and Implementation

1 Objective

Each group must take the Information Retrieval concepts already covered in class and demonstrate how those concepts can be practically applied to the Information Retrieval module of their FYP.

Concept → FYP Problem → Implementation → Retrieval Result → Comparison → Learning

2 Topics to Be Applied

- Boolean Retrieval Model
- Binary Independence Model (BIM)
- BM25 Ranking
- Overlap and Non-Overlap
- Other previously covered IR concepts where relevant



3 Required Work

1 Define the FYP Retrieval Problem

- What information does your FYP need to retrieve?
- What will be the query?
- What will be considered a document/item/result?

2 Apply the IR Technique

Implement at least one appropriate retrieval/ranking technique studied in class.

User Query → IR Model → Matching/Scoring → Ranked Results

3 Show a Working Example

Provide at least 3–5 sample queries from your FYP and demonstrate the retrieved results.

Example:
Query: Best treatment for diabetes
Documents: D1, D2, D3, D4
Apply BM25 → Calculate scores → Rank documents → Show Top Results

D1, D2, D3, D4 → Top Results

1	D2
2	D1
3	D4

4 Explain Using the Richard Feynman Technique

Explain your implementation in very simple language.

"If you cannot explain how your retrieval system selects the best result to a school student, you probably do not understand it well enough yet."

Explain:
What goes in → What happens inside → What comes out → Why that result came first

5 Compare the Result

Where possible, compare at least two retrieval approaches and explain which performs better for your FYP and why.

Examples:
- Boolean Model vs BIM
- BIM vs BM25
- Simple Overlap vs BM25

4 Presentation Requirements



Each group should prepare a concise presentation covering:

1. FYP Problem
2. Retrieval Requirement
3. Selected IR Model
4. Implementation
5. Example Queries
6. Results
7. Comparison
8. Conclusion

Focus on working implementation and understanding, not lengthy theoretical slides.

5 Marking Scheme (10 Marks)

No.	Criteria	Marks
1	Connection of IR concept with FYP	2
2	Correct implementation of IR technique	3
3	Demonstration using queries/results	2
4	Comparison and understanding of retrieval behaviour	2
5	Presentation and Feynman-style explanation	1
	Total	**10**

6 Expected Outcome

By the end of this exercise, students should be able to demonstrate that the IR concepts studied in class are now becoming a **working retrieval component of their FYP.**

Class Concepts → Working Implementation → Measurable Retrieval Performance → Stronger FYP Product

Move your project from Theory → Implementation → Measurable Retrieval Performance → Stronger FYP Product